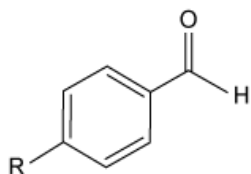


The general structure of a substituted benzaldehyde is shown below. (Note: R = $-\text{OCH}_3$ or $-\text{CH}_3$.)



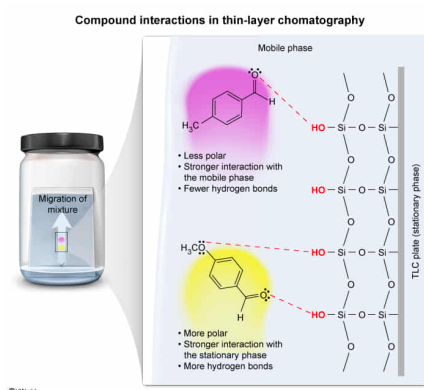
The substituted benzaldehydes are analyzed by thin-layer chromatography on silica plates using 15% ethyl acetate in hexanes. The molecule that has the strongest interaction with the mobile phase is:

- ☐ A. the OCH_3 -substituted benzaldehyde, because it forms more hydrogen bonds than the CH_3 -substituted benzaldehyde. [27%]
- ☐ B. the OCH_3 -substituted benzaldehyde, because it forms fewer hydrogen bonds than the CH_3 -substituted benzaldehyde. [2%]
- ✓ ☒ C. the CH_3 -substituted benzaldehyde, because it forms fewer hydrogen bonds than the OCH_3 -substituted benzaldehyde. [67%]
- ☐ D. the CH_3 -substituted benzaldehyde, because it forms more hydrogen bonds than the OCH_3 -substituted benzaldehyde. [2%]

Correct

68% answered correctly

Explanation:



Thin-layer chromatography (TLC) is a technique used to separate the components of a mixture based on their **polarity**. In normal-phase TLC, the stationary phase is made of a polar, adsorbent material such as silica (SiO_2). The mobile phase is an organic solvent that is much less polar than the stationary phase and travels up the stationary phase via **capillary action**. **Separation** occurs because the mixture components travel up the plate at different rates depending on the **strength** of the competing **intermolecular interactions** between the mixture component and the **stationary and mobile phases**.

In the given benzaldehydes, the partial negative charge (δ^-) on the oxygen atom in the carbonyl ($\text{C}=\text{O}$) acts as a **hydrogen bond acceptor** in both compounds. However, the δ^- on the oxygen atom of the methoxy ($-\text{OCH}_3$) substituent acts as an additional hydrogen bond acceptor whereas the methyl group ($-\text{CH}_3$) is nonpolar and cannot form hydrogen bonds.

Consequently, the OCH_3 -substituted benzaldehyde forms more hydrogen bonds and interacts with the stationary phase to a greater extent (ie, migrates more slowly) than the CH_3 -substituted benzaldehyde, which is less polar and has a stronger interaction with the mobile phase (ie, migrates faster) due to fewer hydrogen bonds formed with the stationary phase.

(Choices A and B) The OCH_3 -substituted benzaldehyde has two hydrogen bond acceptors and forms *more* hydrogen bonds than the CH_3 -substituted benzaldehyde. Therefore, the OCH_3 -substituted benzaldehyde is more polar and has a stronger interaction with the *stationary* phase.

(Choice D) The CH_3 -substituted benzaldehyde forms *fewer* hydrogen bonds than the OCH_3 -substituted benzaldehyde.

Educational objective:

Separation of a mixture using thin-layer chromatography (TLC) occurs because the mixture components travel up the TLC plate at different rates depending on the strength of the intermolecular interactions between the mixture component and the stationary and mobile phases.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical Methods